

R Programming Lab

One-Sample and Two-Sample t-Tests

Program Questions

1. A sample contains the observations

12, 15, 14, 10, 13, 16, 11, 14, 15, 12.

Using a one-sample t-test, test whether the population mean is equal to 12 at the 5% significance level.

2. Using the same data, test whether the population mean is **greater than 12** at the 5% significance level. Use a one-tailed one-sample t-test.
3. Using the same data, construct a 95% confidence interval for the population mean using a one-sample t-test.
4. The marks of eight students before a training programme are

55, 60, 62, 58, 65, 61, 59, 63.

Test whether the mean mark is significantly different from 60 using a one-sample t-test.

5. The marks of two independent groups are given below.

$A = (65, 70, 72, 68, 75, 71, 69, 74)$

$B = (60, 62, 65, 63, 67, 61, 64, 66)$

Perform an independent two-sample t-test to determine whether the two population means are significantly different.

6. Using the data from Question 5, perform a two-sample t-test assuming that the two populations have equal variances.
7. Using the data from Question 5, test whether the mean of Group A is **greater than** the mean of Group B using a one-tailed two-sample t-test.
8. The following marks were obtained from students before and after a training programme.

	1	2	3	4	5	6	7	8
<i>Before</i>	55	60	62	58	65	61	59	63
<i>After</i>	62	65	68	64	70	67	65	69

Perform a paired t-test to determine whether the training programme significantly improved the marks.

9. Create a data frame containing the marks and group labels from Question 5. Use the formula interface `t.test(marks ~ group, data=data)` to perform an independent two-sample t-test.
10. Perform a t-test and extract the following results separately from the R output:
 - (a) t-statistic
 - (b) degrees of freedom
 - (c) p-value
 - (d) confidence interval
 - (e) estimated means

Use a significance level of 5% and state the statistical decision.

Common t-Test Syntaxes in R

Test	Syntax	Example
One-sample t-test	<code>t.test(x, mu = mu0)</code>	<code>t.test(x, mu = 12)</code>
One-sample: greater	<code>t.test(x, mu = mu0,</code> <code>alternative = "greater")</code>	
One-sample: less	<code>t.test(x, mu = mu0,</code> <code>alternative = "less")</code>	
Confidence interval	<code>t.test(x, mu = mu0,</code> 95% CI for population mean <code>conf.level = 0.95)</code>	
Independent two-sample	<code>t.test(x, y)</code>	<code>t.test(groupA, groupB)</code>
Equal variance	<code>t.test(x, y, var.equal = TRUE)</code> <code>var.equal = TRUE)</code>	<code>t.test(groupA, groupB,</code>
Two-sample: greater	<code>t.test(x, y,</code> Tests whether $\mu_x > \mu_y$ <code>alternative = "greater")</code>	

Test	Syntax	Example
Two-sample: less alternative = "less")	<code>t.test(x, y,</code> Tests whether $\mu_x < \mu_y$ <code>)</code>	
Paired t-test	<code>t.test(x, y, paired = TRUE)</code> paired = TRUE)	<code>t.test(before, after,</code> <code>)</code>
Formula interface data = data)	<code>t.test(y ~ group, data = df)</code> data = data)	<code>t.test(marks ~ group,</code> <code>)</code>
Store result	<code>result <- t.test(x, mu = 12)</code>	Stores the complete test result
Extract t-statistic	<code>result\$statistic</code>	Returns calculated t-value
Extract p-value	<code>result\$p.value</code>	Returns p-value
Extract confidence interval	<code>result\$conf.int</code>	Returns confidence interval

Important Arguments

- `mu`: hypothesized population mean.
- `alternative = "two.sided"`: default two-tailed test.
- `alternative = "greater"`: right-tailed test.
- `alternative = "less"`: left-tailed test.
- `paired = TRUE`: paired t-test.
- `var.equal = TRUE`: assumes equal population variances.
- `conf.level = 0.95`: specifies a 95% confidence level.